COMSATS University Islamabad, Lahore Campus

Terminal, Lab Examination – Spring 2025

Course Title:	Web Technologies			Course Code:	CSC336	Credit	3(2,1)
Course Instructor/s:	Asif Shahzad			Programme Name:	BSCS		
Semester:	6	Batch: FA22	Section:	G2	Date:	Jun. 17, 2025	
Time Allowed:	**2.5 Hours**			**Maximum Marks:**	**50**		
Student's Name:				Reg. No.			

Question 1: [50 Marks]

Create a web application with the following features. Use Bootstrap for tables, forms, buttons, etc. The app should function entirely from the homepage (`http://localhost:3000/`), means, without requiring user to enter any other URL in address bar. After each successful operation, redirect user to homepage.

(a) Create a `users.json` file with username and password pairs. Only listed users can log in. Add a **Login** link in the header to open the login form. Show relevant error messages (e.g., invalid username/password) using sessions. After login, replace **Login** with a **Logout** link to log the user out. [10 Marks]

(b) Create a `products.json` file containing an array of at least 4 product objects with `id`, `title`, `price`, and `picture_filename`. On the home page, display all products in a table. If the user is logged in, show an **Edit** link for each product. Clicking it opens a pre-filled update form. On submit, update the `products.json` file with the new details including image. [10 Marks]

(c) On the homepage, show a **Delete** link in each product row **only for logged-in users**. On click, use **AJAX** to remove the product from `products.json` and **immediately remove** the row from the browser without refreshing the page. [10 Marks]

(d) For logged-in users, display an **"Add to Cart"** button with a quantity field (default: 1) for each product. On click, add the product to the server-side session cart. Preserve the cart across logins by restoring previous items added in cart. In the header, show a **"View Cart"** link displaying all items, quantities, total bill, and a **"Delete"** link to remove individual item from cart. [20 Marks]